DM RAFIUN BIN MASUD

AI Engineer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience turning AI concepts into production-ready applications. Skilled in Python, TensorFlow, PyTorch, Scikit-learn, OpenCV, and FastAPI, with experience across the machine learning lifecycle, including data preprocessing, model development, API integration, and Dockerized deployment. Particularly interested in Large Language Models, Retrieval-Augmented Generation (RAG), AI Agents, Computer Vision, and MLOps, with a strong focus on combining practical engineering with applied AI research. Author of peer-reviewed publications (Springer, IEEE), including a Best Paper Award at ICIDA 2022.

WORK EXPERIENCE

AI Engineer – SM Technology March 2025 – Present

Dhaka, Bangladesh

- **AI Backend Development:** Design and develop scalable AI backend systems using Python and FastAPI to power intelligent web and mobile applications.
- **LLM & RAG Systems:** Work across Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Computer Vision, and AI-driven automation.
- **Production Readiness:** Build robust, maintainable AI solutions with an emphasis on real-world deployment.

Research Assistant – East West University Jan 2026 – Present

Dhaka, Bangladesh

- **Capstone Project:** Completed a four-member capstone project under the supervision of Dr. Md. Sawkat Ali, Associate Professor.
- **Project Leadership:** Led project planning, task allocation, stakeholder communication, and progress reporting to ensure timely delivery.

IT Support Engineer – Taltola Dot Net March 2021 – January 2022

Dhaka, Bangladesh

- **Network Operations:** Managed end-to-end network operations, including configuration, installation, maintenance, and upgrades.
- **Proactive Monitoring:** Monitored IT systems to detect and resolve performance bottlenecks before they impacted users.

PROJECTS

- **Apple Plum Growth Stage Classification** (TensorFlow, Keras, OpenCV, Streamlit, 2022): Curated a 701-image agricultural dataset and engineered preprocessing pipelines and CNN architectures for growth-stage classification and bounding-box detection.
- **MANIFEX AI Paragraph Evaluator** (NLP, LLM, Docker – manifex.org): Built an NLP engine that evaluates sentence structure, grammar, vocabulary richness, and overall writing quality with automated scoring.
- **Piper TTS FastAPI Inference Server** (FastAPI, Docker, REST API – GitHub): Built a low-latency, containerized neural text-to-speech server delivering real-time voice synthesis via high-throughput REST endpoints.
- **Movie Recommendation Engine** (Scikit-learn, Collaborative Filtering – GitHub): Built a dual-engine recommender combining content-based cosine similarity and collaborative-filtering matrix factorization on large user-rating matrices.
- **DFT Image Prep & Enhancement** (GitHub): Super-resolution and background removal pipeline for image enhancement.
- **Bilingual Bengali Medicine-Recommendation Chatbot** (Python, LLMs – GitHub): Chatbot supporting Bengali for medicine information retrieval.

RESEARCH PUBLICATIONS

- **Deep Learning and Non-Contrast CT for Widespread Pancreatic Cancer Detection** – *Best Paper Award*, International Conference on Innovations in Data Analytics (Springer ICIDA), 2022. Designed and trained CNNs to detect pancreatic cancer from non-contrast CT scans, improving early screening efficacy without contrast agents. DOI: 10.1007/978-981-96-6303-3_38
- **ShrimpXNet: A Transfer Learning Framework for Shrimp Disease Classification with Augmented Regularization, Adversarial Training, and Explainable AI** – IEEE, 2025. Built a transfer learning vision framework for aquaculture disease classification, using Grad-CAM for interpretability. DOI: 10.1109/3ict68299.2025.11442114
- **SFiNNet: Leveraging Transfer Learning and Explainable AI for Automated Small Fish Classification** – IEEE, 2025. Proposed a species-recognition model combining transfer learning, explainable AI, and adversarial training for small-sample marine datasets. DOI: 10.1109/ICCIT68739.2025.11491764

- **Emerging Trends in Data Leakage Detection and Prevention: A Comprehensive Review** – Springer, 2026. DOI: 10.1007/978-3-032-10386-4_38

SKILLS

- **Programming Languages:** Python, C++, C, Java, PHP, SQL
- **AI/ML & Generative AI:** TensorFlow, PyTorch, Keras, Scikit-learn, Transformers, Hugging Face, NLTK, NLP, Large Language Models (LLMs), RAG, LangGraph, Explainable AI (XAI), Transfer Learning
- **Computer Vision & Speech:** OpenCV, Image Classification, Object Detection, Data Augmentation, Adversarial Training, Speech Recognition, Audio Processing
- **Backend & APIs:** FastAPI, REST APIs, Async Processing, Microservices, Postman
- **Databases & Vector Stores:** MySQL, Oracle SQL, Vector Databases
- **Cloud & DevOps:** Docker, Git, GitHub, GitHub Actions, CI/CD, nginx, AWS, GCP, VPS Deployment
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Extraction, Statistical Analysis
- **Web Technologies:** React, HTML, CSS

EDUCATION

East West University Dhaka, Bangladesh
Bachelor of Science in Computer Science & Engineering – CGPA: 3.05 / 4.00 *2024*

CERTIFICATIONS

Data Analytics and Visualization Job Simulation – Accenture North America (via Forage), 2024

LEADERSHIP & COMMUNITY ENGAGEMENT

Volunteer – IEEE Computer Society Student Branch, East West University Jan 2022 – Aug 2023

- Coordinated and organized technical events, workshops, and student activities to ensure smooth execution.

PROGRAMMING ACTIVITIES

- **Codeforces:** Participated in multiple competitive programming contests.
- **beecrowd:** Solved 100+ programming problems.

LANGUAGES

Bengali – Native English – Fluent

REFERENCES

Dr. Md Sawkat Ali (Supervisor)
Associate Professor, Dept. of CSE, East West University
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Dr. Mohammad Rifat Ahmmad Rashid (Faculty)
Associate Professor, Dept. of CSE, East West University
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